



Tribhuvan University

Faculty of Humanities and Social Science

A Project Report On

**Secure Health Information Exchange System with Consent-Based and
Emergency Access Control**

Submitted to:

Department of Computer Application

Pascal National College

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Submitted by:

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Under the Supervisor

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1. Introduction

In today's rapidly evolving healthcare sector, timely access to accurate and complete patient medical records is critical for delivering quality care. However, patient health information in Nepal is largely fragmented and stored in isolated systems across different hospitals. This lack of interoperability often leads to delays in treatment, repeated medical tests, and increased risk of medical errors, especially when patients seek care from multiple healthcare providers.

A Secure Health Information Exchange (HIE) System is a specialized digital platform that enables hospitals to request and share patient medical records in a secure, controlled, and patient-centric manner. Unlike traditional methods such as fax, email, or manual document transfer — which are slow, insecure, and offer little patient control — this system provides a structured and technology-driven solution.

The core principle of the proposed system is “Patient owns and controls their own health data.” Any sharing of medical records requires dual consent — explicit approval from both the patient (data owner) and the data-holding hospital (data custodian). The system also incorporates a strictly controlled emergency access mechanism that allows limited access to critical information (such as blood group, allergies, and current medications) for a maximum of 24 hours, with mandatory justification and full audit trail.

Every action performed in the system — whether requesting records, granting consent, or accessing data — is comprehensively logged in an audit trail to ensure transparency, accountability, and non-repudiation. By leveraging modern web technologies such as React.js, Node.js, and PostgreSQL with robust authentication and authorization mechanisms, the proposed Secure Health Information Exchange System aims to bridge the gap between isolated hospital systems in Nepal.

This project seeks to enhance coordination among healthcare institutions, reduce delays in emergency situations, strengthen patient privacy, and ultimately contribute to improved healthcare outcomes. Proposed HIE system aims to build trust among patients, doctors, and healthcare institutions.

2. Problem Statement

The healthcare system in Nepal continues to face significant challenges due to fragmented and isolated medical record management. Most hospitals, especially in government and rural areas, maintain patient records in paper-based or standalone digital systems that do not communicate with each other. As a result, when patients visit different hospitals, their complete medical history is often unavailable, leading to repeated diagnostic tests, delayed treatment, and increased risk of medical errors.

In emergency situations, the lack of quick access to critical information such as blood group, allergies, chronic conditions, and current medications can have serious, sometimes life-threatening consequences. Furthermore, there is no standardized, secure, and efficient mechanism for hospitals to request and share patient medical records across institutions while respecting patient privacy and consent.

Existing systems either completely ignore patient consent or provide overly complex solutions that are difficult and expensive for small and medium-sized hospitals to adopt. Issues such as unauthorized access, lack of accountability, and weak audit mechanisms further increase the risk of data breaches and misuse of sensitive health information.

Therefore, there is a pressing need for a secure, patient-centric Health Information Exchange (HIE) system that enables controlled sharing of medical records between hospitals through **dual consent** (patient and holding hospital), supports emergency access with strict controls, and maintains complete auditability of all activities.

3. Objective

The main objectives of the proposed Secure Health Information Exchange System are as follows:

- To design and develop a secure web-based platform for exchanging patient medical records between different hospitals.
- To implement a dual consent mechanism where both the patient and the data-holding hospital must approve before any record sharing occurs.
- To provide a strictly controlled emergency access feature that grants time-limited access to critical patient data with proper justification and auditing.
- To ensure patient data ownership and privacy through strong authentication, authorization, and role-based access control.
- To maintain comprehensive audit logs for all system activities to ensure transparency and accountability.
- To develop a user-friendly interface for patients, doctors, and hospital administrators.
- To demonstrate the feasibility of secure health data exchange in the Nepalese healthcare context using modern technologies.

4. Methodology

Agile model will be implemented to carry on this project with additional development and revision phase as smaller pieces of work will be carried out regularly instead of one big launch to adapt to changes quickly.

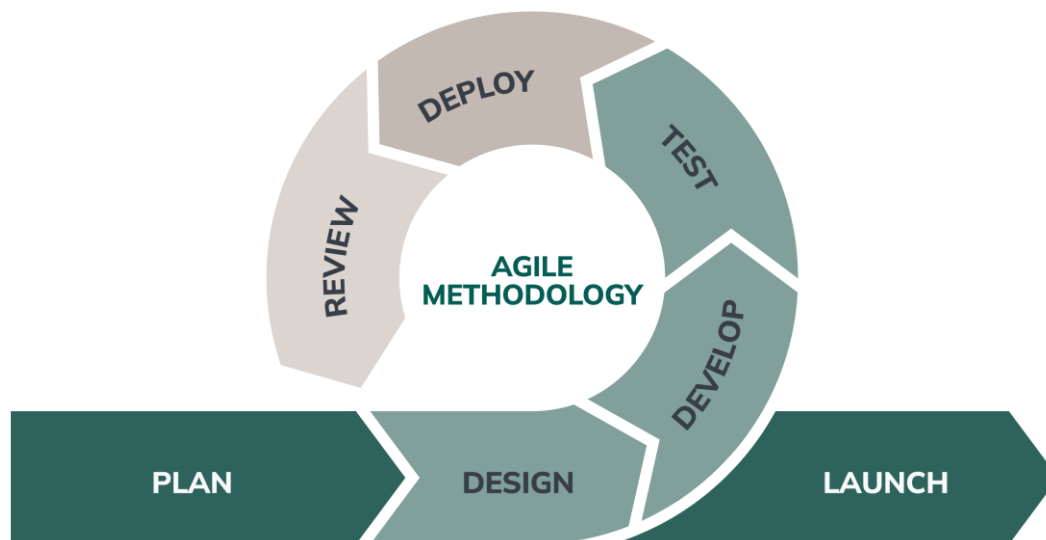


Figure 1: Agile Methodology

a. Study of existing System

The healthcare system in Nepal consists of both public and private institutions serving a rapidly growing population. However, most hospitals still maintain patient medical records in isolated paper-based or standalone digital systems. This fragmented approach leads to poor interoperability between healthcare facilities [1].

When patients visit different hospitals — which is very common due to referrals, specialist care, or personal preference — their complete medical history is often unavailable to the treating doctors. As a result, patients frequently undergo repeated laboratory tests and diagnostic procedures, leading to increased healthcare costs, delays in treatment, and higher chances of medical errors. In emergency situations, the lack of immediate access to critical information such as blood group, allergies, chronic conditions, and current medications can have serious, sometimes life-threatening consequences [2].

With the advancement of information and communication technology and increasing smartphone penetration, the Government of Nepal has recognized the importance of digital health and has introduced the Digital Health Strategy to promote electronic health records and

system interoperability [3]. Despite these efforts, the implementation of a secure and standardized Health Information Exchange (HIE) system remains limited, especially in rural and government hospitals.

This project is motivated by the need to address these challenges by developing a Secure Health Information Exchange (HIE) System that enables controlled sharing of patient medical records while ensuring patient ownership, dual consent, and secure emergency access.

b. Literature Review

Health Information Exchange (HIE) has been extensively studied worldwide as a promising solution to improve healthcare coordination and reduce medical errors. Haux [4] highlighted the evolution of health information systems from isolated departmental solutions toward integrated platforms. Similarly, Adler-Milstein and Jha [5] found that hospitals with effective HIE systems demonstrated significantly better information sharing and improved patient outcomes.

In recent years, research focus has shifted toward patient-centric approaches. Zhang et al. [6] proposed secure and efficient data sharing frameworks for cloud-based healthcare systems using cryptographic techniques. Several studies have emphasized the importance of patient consent mechanisms and emergency access controls to balance privacy protection with the need for urgent medical care [7].

In the context of developing countries, researchers have identified major challenges including limited digital infrastructure, lack of data standardization, low digital literacy, and resource constraints [8]. While many global HIE platforms exist, most of them are either too complex and expensive for small hospitals or do not adequately empower patients with control over their own health data.

Although considerable progress has been made in Health Information Exchange systems globally, there is still a significant gap in lightweight, consent-driven, and emergency-enabled solutions suitable for resource-limited settings like Nepal. This project aims to bridge this gap by developing a simplified yet secure HIE prototype that focuses on dual consent, time-bound emergency access, and comprehensive audit logging.

c. Requirement Analysis

I. Functional Requirement

- User (patients, doctor, hospitals) can register and log in securely
- Hospitals can request and share patient medical records
- Patients can approve or reject data access requests
- System provides emergency access to limited critical data

II. Non-Functional Requirements

- System provides emergency access to limited critical data
- System should have fast response time and good performance
- System must be easy to use (user-friendly interface)
- System should be reliable and scalable

d. Feasibility Study

a. Technical Feasibility:

The system can be built using React.js(frontend), Node.js(backend) and PostgreSQL(database), which are widely used and well-supported. These technologies support scalable web applications and secure data handling. Availability of tools, libraries, and documentation makes development practical within the project timeframe. Integration using APIs (Node.js + PostgreSQL via ORM) is straightforward and reliable.

b. Operational Feasibility:

The system ensures easy to use for patients, doctors and hospitals with basics training. The system improves workflow by providing quick access to patient medical records and helps to make better decision-making, especially in emergency situations. The system adoption may require initial training and willingness for hospitals.

c. Economic Feasibility:

Development of this system is highly economically feasible because of open-source. The system may cost minimal cost for development (can use low-cost or free hosting for prototype). This system reduces manual paperwork and improves efficiency, leading to long-term benefits.

d. High level Design of system

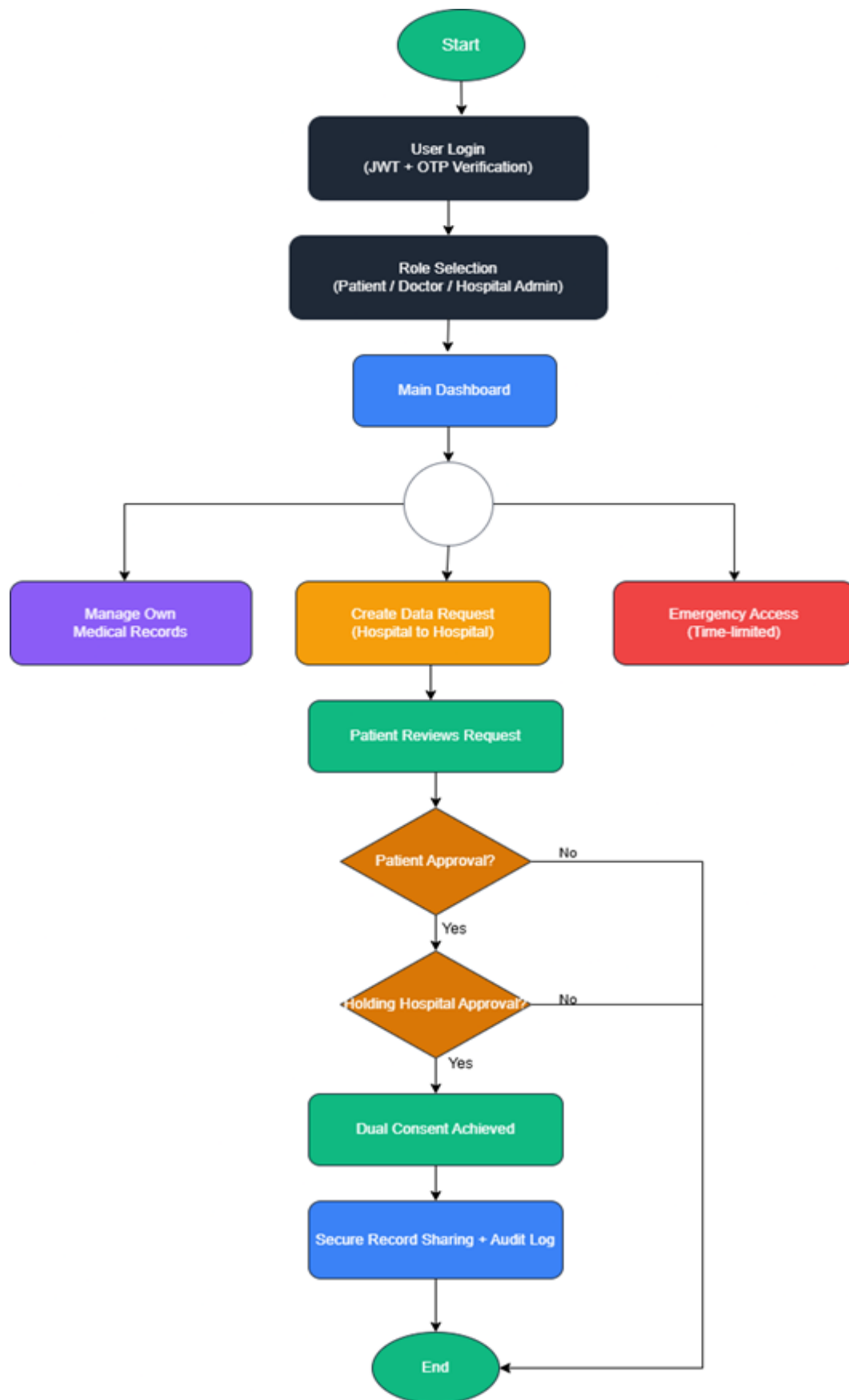


Figure 2: Flow chart of secure Health information System

Working Mechanism of the System

The Secure Health Information Exchange System works through a well-defined sequence of steps that ensure secure, controlled, and auditable sharing of patient medical records between different hospitals. Initially, the user logs into the system using secure authentication with JWT and OTP verification. Based on their role — Patient, Doctor, or Hospital Admin — they are directed to the respective dashboard.

When a doctor needs access to a patient's medical record stored in another hospital, they create a Data Request through the system. The request is then sent to the patient for review. The patient can approve or reject the request after viewing the details such as the reason and the requesting doctor's information. If the patient approves the request, it is forwarded to the data-holding hospital for final approval.

Only after receiving dual consent from both the patient and the holding hospital, the system allows secure sharing of the requested medical records. In emergency situations, the system provides a strictly controlled emergency access mechanism. An authorized doctor can request temporary access to critical patient data (such as blood group, allergies, current medications, and chronic conditions) by providing a valid justification. The system grants limited access for a maximum of 24 hours and automatically revokes it after the expiry time.

Throughout the entire process, every action — including login, request creation, consent approval, record viewing, and emergency access — is recorded in the Audit Log for transparency and accountability. The system ensures patient data ownership by giving patients full control over who can access their medical records.

Finally, all shared records, consents, and access activities are properly tracked, and notifications are sent to the relevant users, making the process transparent, secure, and user-friendly.

Algorithm Detail:

Dual Consent Management Algorithm:

Dual Consent Management Algorithm is the core algorithm used to ensure secure and controlled sharing of patient medical records across different hospitals. In this project, it is used to determine whether a medical record can be shared only after receiving mandatory approval from both the patient (data owner) and the holding hospital (data custodian).

This algorithm follows a sequential approval workflow. The result ranges from “pending” to “approved” or “rejected”. A higher approval rate (both consents received) indicates that the record can be safely shared. This algorithm helps in maintaining patient privacy, data ownership, and regulatory compliance while making it easier for doctors to access necessary records with proper authorization.

Formula (Conceptual State Transition): $\text{Status} = f(\text{PatientApproval} \wedge \text{HospitalApproval})$

Algorithm steps:

1. A doctor from the requesting hospital creates a new `DataRequest` for a patient’s medical record located in another hospital.
2. The system notifies the patient with request details (reason, requesting doctor, and hospital name).
3. The patient reviews the request and decides whether to **Approve** or **Reject** it.
4. If the patient rejects the request, the status is updated to “rejected” and the process ends.
5. If the patient approves, the request status is changed to “patient_approved” and the system forwards the request to the holding hospital.
6. The holding hospital (admin or authorized doctor) reviews the request and decides to **Approve** or **Reject**.
7. If the holding hospital approves and the patient has already approved, the system changes the status to “approved”.
8. The system then creates `SharedRecord` entries, allowing secure access to the medical records.
9. Audit logs are generated at every step for full traceability.
10. Notifications are sent to the patient and the requesting doctor.

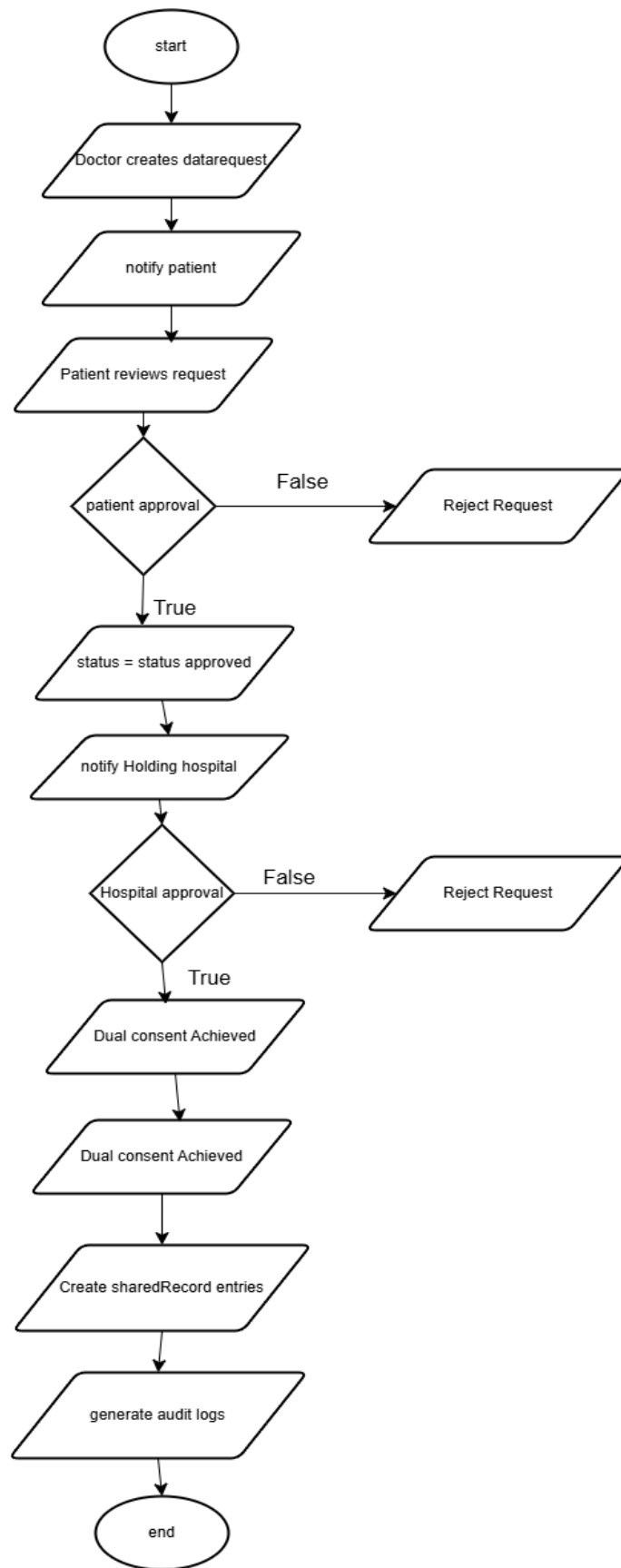


Figure 3: Dual Consent Management Algorithm

Emergency Access Control Algorithm:

Emergency Access Control Algorithm is a time-bound algorithm used to grant temporary access to critical patient health information during medical emergencies. In this project, it is used to allow limited access to vital data such as blood group, allergies, current medications, and chronic conditions while maintaining strict control and full auditing.

This algorithm calculates whether the access request is valid and within the allowed time window. The result is either “granted” (with expiry) or “denied”. It ensures that emergency access is provided only when justified and automatically revoked after a maximum of 24 hours. This algorithm helps in saving lives during critical situations while protecting patient privacy through proper logging and post-access notification.

Key Rule:

$\text{Access} = \text{ValidDoctor} \wedge \text{JustificationProvided} \wedge \text{CurrentTime} < \text{expiresAt}$

Algorithm steps:

1. An authorized doctor submits an emergency access request along with a valid justification (reason).
2. The system validates the doctor’s role, hospital affiliation, and the provided justification.
3. If validation fails, the request is denied and logged.
4. If valid, the system grants access to limited critical data only.
5. The system sets an expiry time (expiresAt) — default is 24 hours from the grant time.
6. All access activities are recorded in the EmergencyAccess and AuditLog tables.
7. The patient is notified about the emergency access (immediately or after the event).
8. After the expiresAt time passes, the system automatically revokes the access.
9. Any attempt to access data after expiry is denied by the system.

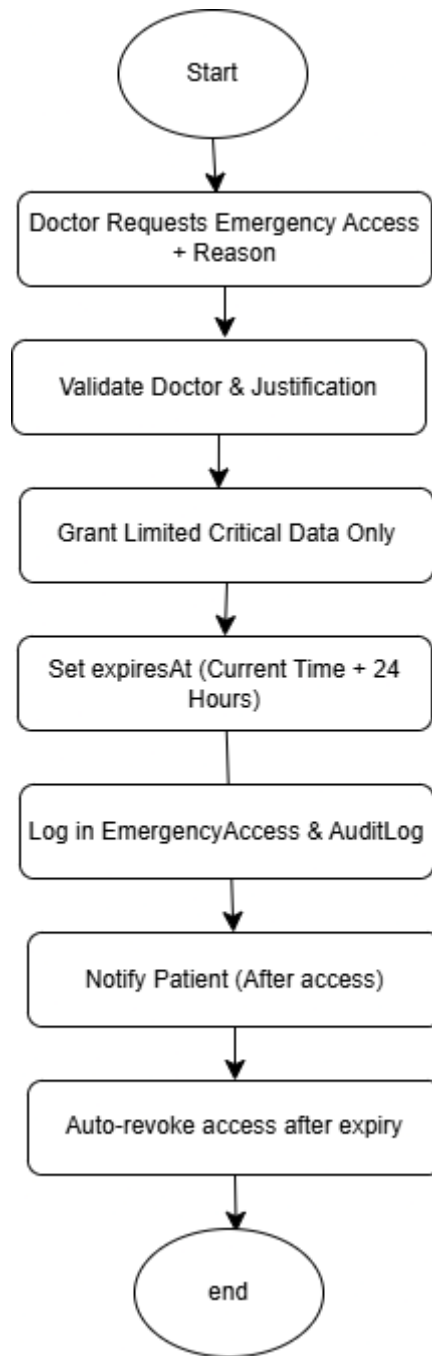


Figure 4: Emergency Access control

Audit Logging Algorithm:

Audit Logging Algorithm is a systematic algorithm used to record every action performed in the system for security, accountability, and compliance purposes. In this project, it is used to track all sensitive operations such as creating data requests, approving consents, viewing records, granting emergency access, and sharing medical records

This algorithm captures essential details (who, what, when, and why) and stores them immutably. The result is a complete, tamper-evident history of system activities. This algorithm helps hospitals and patients maintain transparency and supports forensic analysis if needed.

Algorithm Steps:

- Capture action details (userId, action type, entity, metadata, IP address)
- Create AuditLog entry
- Store record in database (append-only)

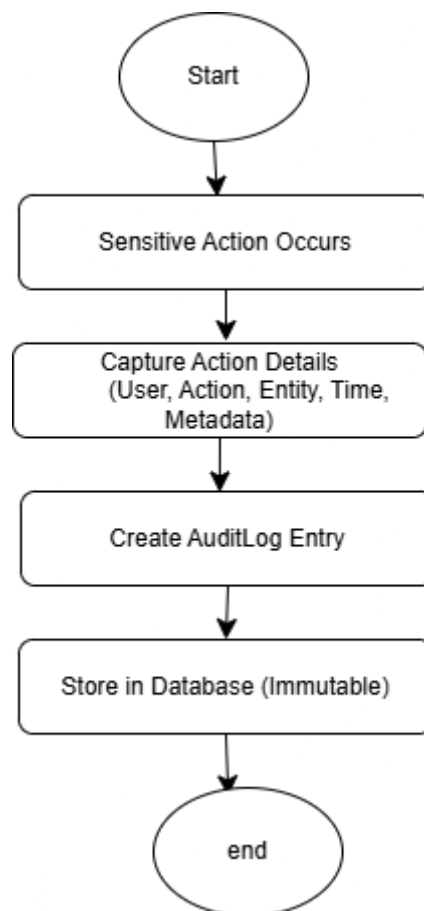


Figure 5: Audit logging algorithm

Data Request Conflict Detection Algorithm:

Data Request Conflict Detection Algorithm is used to prevent duplicate or overlapping data sharing requests between hospitals. In this project, it checks whether a similar request for the same patient record between the same pair of hospitals is already pending or active.

This algorithm helps avoid redundancy and confusion in the consent workflow. If a conflicting request exists, the new request is rejected with a proper message.

Algorithm steps:

1. When a doctor attempts to create a new DataRequest, the system collects key parameters (patientId, fromHospitalId, and toHospitalId).
2. The system queries the database to check if any existing request already exists with the same combination of patient and hospitals.
3. If an active or pending request is found (status = pending, patient_approved, or approved), the system rejects the new request.
4. The user is shown a proper message: “A similar request is already in progress.”
5. If no conflicting active request is found, or if the previous request was rejected or expired, the system allows the creation of the new DataRequest.
6. The result of the conflict check is recorded in the AuditLog table for traceability.

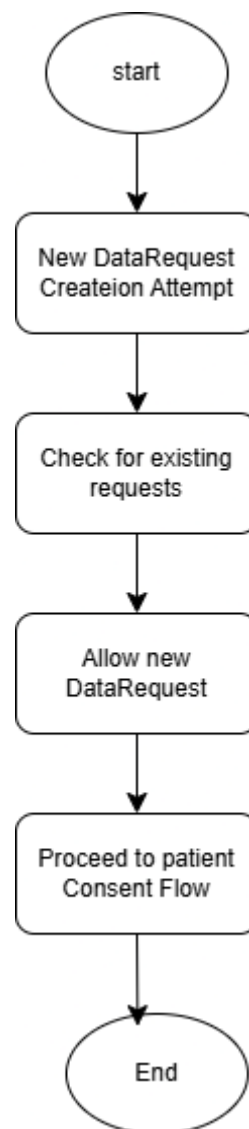


Figure 6: Data confilct Detection

Record Expiry and Cleanup Algorithm:

Record Expiry and Cleanup Algorithm is a background maintenance algorithm used to automatically remove or archive expired temporary data such as OTPs and emergency access permissions. In this project, it runs periodically to maintain system security and performance.

This algorithm queries for records where the expiry time has passed and safely cleans them up while logging the activity.

Algorithm steps:

1. A scheduled background job (cron job) runs periodically (e.g., every hour or daily).
2. The system queries the database for expired records:
 - a. OTPs where $\text{expiresAt} < \text{current time}$ or $\text{isUsed} = \text{true}$
 - b. Emergency access records where $\text{expiresAt} < \text{current time}$
3. Expired OTP records are marked as used or deleted from the system
4. Expired Emergency Access entries are revoked or archived.

The cleanup activity is logged in the AuditLog table with details of records processed

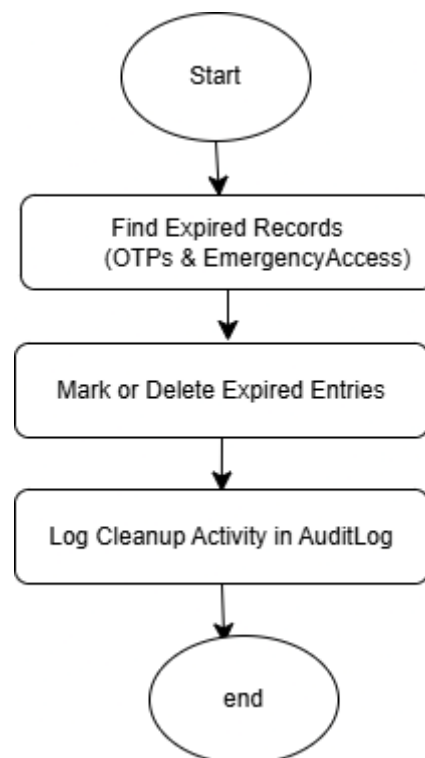


Figure 7: Record Expiry and cleanup

5. Gantt Chart

	Start Date	End Date	1	2	3	4	5	6	7	8	Status
Research and Planning	15-April	19-April									Completed
Requirement Gathering	20-April	25-April									Ongoing
System Design	26-April	10-May									Upcoming
System Development	11-May	20-May									Upcoming
Review	21-May	30-May									Upcoming
Testing	1-June	8-June									Upcoming

6. Expected outcome

The expected outcome of this project is the development of a fully functional Secure Health Information Exchange System with consent-based and emergency access control. The system will allow hospitals to securely request and share patient medical records only after receiving dual approval from both the patient and the data-holding hospital. It will also provide a strictly controlled emergency access mechanism that grants time-limited access to critical patient information with proper justification and full auditing.

The system will ensure patient data ownership and privacy through strong authentication, authorization, and role-based access control, while maintaining comprehensive audit logs for all activities to ensure transparency and accountability. A user-friendly interface will be developed for patients, doctors, and hospital administrators. Overall, the project will successfully demonstrate the feasibility of secure health data exchange in the Nepalese healthcare context using modern web technologies.

7. References

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